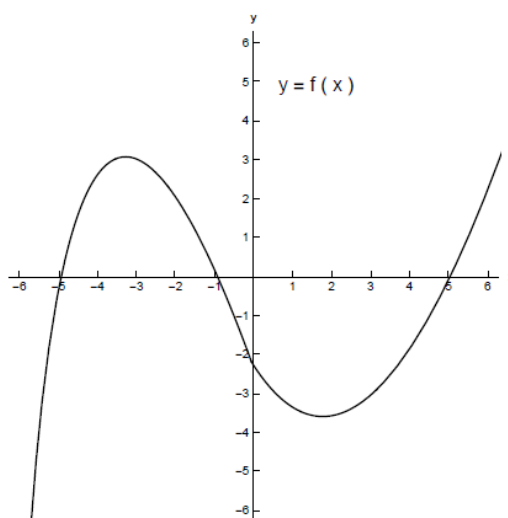


Problem 5

Problem 5

Problem 1 Suppose we are given the graph of a function f :



- (a) Use this graph to find the following: (Assume all values will be integers or $+\infty$ or $-\infty$)
- (i) all x where $f(x) = 0$,
 - (ii) all x where $f(x) > 0$,
 - (iii) all x where $f(x) < 0$, and
 - (iv) all x where $f(x)$ attains a local maximum and all x where f attains a local minimum.

Without sketching the graph of f' find

- (b) (i) all x where $f'(x) = 0$,
- (ii) all x where $f'(x) > 0$,
- (iii) all x where $f'(x) < 0$, and
- (iv) On the following intervals, does $f'(x)$ seem to be increasing or decreasing?
- i. $(-\infty, 0)$
 - ii. $(0, \infty)$

Problem 5

- (c) *Sketch a graph of f' .*

[2]